

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T016	Name: Himani Malankar
Class:TYIT	Batch:1
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A. Develop a PWA that works in offline mode using Service Worker.

CODE:

(Offline.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Offline</title>

  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: Arial, Helvetica, sans-serif;
    }

    body {
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      background: linear-gradient(135deg,#0d6efd,#5b9cff);
    }

    .card {
      background: #fff;
      padding: 40px;
      width: 380px;
      border-radius: 15px;
      text-align: center;
      box-shadow: 0 8px 20px rgba(0,0,0,0.2);
    }

    .icon {
      font-size: 70px;
      margin-bottom: 15px;
    }

    h1 {
      color: #0d6efd;
      margin-bottom: 10px;
    }
  </style>
</html>
```

```

p {
  color: #555;
  line-height: 1.6;
  margin-bottom: 25px;
}

button {
  background: #0d6efd;
  color: white;
  border: none;
  padding: 12px 30px;
  font-size: 16px;
  border-radius: 8px;
  cursor: pointer;
  transition: 0.3s;
}

button:hover {
  background: #0b5ed7;
}
</style>
</head>
<body>

<div class="card">
  <div class="icon"><img alt="Offline icon" data-bbox="358 413 378 428"/></div>

  <h1>You're Offline</h1>

  <p>
    Internet connection is unavailable.<br>
    Please check your network and try again.
  </p>

  <button onclick="location.reload()">
    Retry
  </button>
</div>

</body>
</html>

```

(_Layout.cshtml):

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - Practical 7A</title>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/Practical 7A.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json" />
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white
border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">Practical 7A</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"

```

```

        aria-expanded="false" aria-label="Toggle navigation">
        <span class="navbar-toggler-icon"></span>
    </button>
    <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
        <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
                <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
            </li>
            <li class="nav-item">
                <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
            </li>
        </ul>
    </div>
</div>
</nav>
</header>
<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - Practical 7A - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
    </div>
</footer>
<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
<script src="~/js/site.js" asp-append-version="true"></script>
@await RenderSectionAsync("Scripts", required: false)

<script>if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js');
}
</script>

</body>
</html>

```

(service-worker.js):

```

const CACHE = "v1";

self.addEventListener("install", event => {

    console.log("Service Worker Installed");

    event.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/index.html"
            ]);
        })
    );
});

self.addEventListener("fetch", event => {

    event.respondWith(
        fetch(event.request)
    );
});

```

```
        .catch(() => caches.match("/index.html"))
    );
});
(manifest.json):
```

```
{
  "name": "College Information Portal",
  "short_name": "PAI",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",

  "icons": [
    {
      "src": "/Icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "/Icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

OUTPUT:

